

**SAFETY DATA SHEET**

Creation Date 21-September-2012

Revision Date 26-March-2024

Revision Number 3

**1. Identification**

**Product Name** (+/-)-Styrene oxide

**Cat No. :** L07821

**CAS-No** 96-09-3  
**Synonyms** 1,2-Epoxyethylbenzene

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

**Details of the supplier of the safety data sheet****Company****Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

**Emergency Telephone Number**For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887**2. Hazard(s) identification****Classification****WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                   |             |
|-----------------------------------|-------------|
| Flammable liquids                 | Category 4  |
| Acute dermal toxicity             | Category 4  |
| Acute Inhalation Toxicity         | Category 3  |
| Skin Corrosion/Irritation         | Category 2  |
| Serious Eye Damage/Eye Irritation | Category 2  |
| Skin Sensitization                | Category 1  |
| Germ Cell Mutagenicity            | Category 1B |
| Carcinogenicity                   | Category 1B |

**Label Elements****Signal Word**

Danger

**Hazard Statements**

Combustible liquid  
Harmful in contact with skin  
Toxic if inhaled  
Causes skin irritation  
May cause an allergic skin reaction  
Causes serious eye irritation  
May cause genetic defects  
May cause cancer

**Precautionary Statements****Prevention**

Obtain special instructions before use  
Do not handle until all safety precautions have been read and understood  
Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
Avoid breathing dust/fume/gas/mist/vapors/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Use only outdoors or in a well-ventilated area  
Contaminated work clothing should not be allowed out of the workplace  
Wear protective gloves/protective clothing/eye protection/face protection

**Response**

IF ON SKIN: Wash with plenty of soap and water  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
IF exposed or concerned: Get medical advice/attention  
Call a POISON CENTER/ doctor  
In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
Take off contaminated clothing and wash it before reuse

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component        | CAS-No  | Weight % |
|------------------|---------|----------|
| Oxirane, phenyl- | 96-09-3 | >95      |

### 4. First-aid measures

**General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

**Eye Contact**

In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

**Skin Contact**

Immediate medical attention is required. Wash off immediately with plenty of water for at least 15 minutes.

|                                        |                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                      | Remove to fresh air. If breathing is difficult, give oxygen. Immediate medical attention is required. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device.                 |
| <b>Ingestion</b>                       | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                          |
| <b>Most important symptoms/effects</b> | Difficulty in breathing. . May cause allergic skin reaction. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                                                                   |

## 5. Fire-fighting measures

|                                         |                                                                                                                                         |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray, carbon dioxide (CO <sub>2</sub> ), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                                                |
| <b>Flash Point</b>                      | 74 °C / 165.2 °F                                                                                                                        |
| <b>Method -</b>                         | No information available                                                                                                                |
| <b>Autoignition Temperature</b>         | 498 °C / 928.4 °F                                                                                                                       |
| <b>Explosion Limits</b>                 |                                                                                                                                         |
| <b>Upper</b>                            | 22.00%                                                                                                                                  |
| <b>Lower</b>                            | 1.10%                                                                                                                                   |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                                |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                                |

### Specific Hazards Arising from the Chemical

Combustible material. Containers may explode when heated. Keep product and empty container away from heat and sources of ignition. Risk of ignition.

### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

|               |                     |                    |                         |
|---------------|---------------------|--------------------|-------------------------|
| <b>Health</b> | <b>Flammability</b> | <b>Instability</b> | <b>Physical hazards</b> |
| 3             | 2                   | 0                  | N/A                     |

## 6. Accidental release measures

|                                             |                                                                                                                                                                                                                                                      |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges. |
| <b>Environmental Precautions</b>            | Should not be released into the environment. See Section 12 for additional Ecological Information.                                                                                                                                                   |
| <b>Methods for Containment and Clean Up</b> | Keep in suitable, closed containers for disposal. Soak up with inert absorbent material. Remove all sources of ignition.                                                                                                                             |

## 7. Handling and storage

**Handling** Do not breathe mist/vapors/spray. Do not get in eyes, on skin, or on clothing. Do not ingest. If swallowed then seek immediate medical assistance. Use only under a chemical fume hood. Wear personal protective equipment/face protection. Keep away from open flames, hot surfaces and sources of ignition.

**Storage.** Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. To maintain product quality: Keep refrigerated. Incompatible Materials. Strong oxidizing agents.

## 8. Exposure controls / personal protection

**Exposure Guidelines** This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

| Component        | Alberta | British Columbia | Ontario TWAEV | Quebec | ACGIH TLV          | OSHA PEL | NIOSH |
|------------------|---------|------------------|---------------|--------|--------------------|----------|-------|
| Oxirane, phenyl- |         | Skin             |               |        | TWA: 1 ppm<br>Skin |          |       |

**Engineering Measures** Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

**Eye Protection**  
**Hand Protection**

Goggles  
Protective gloves

| Glove material                                      | Breakthrough time                 | Glove thickness | Glove comments         |
|-----------------------------------------------------|-----------------------------------|-----------------|------------------------|
| Nitrile rubber<br>Neoprene<br>Natural rubber<br>PVC | See manufacturers recommendations | -               | Splash protection only |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

When RPE is used a face piece Fit Test should be conducted

### Environmental exposure controls

No information available.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

**Physical State**

Liquid

|                                        |                              |
|----------------------------------------|------------------------------|
| Appearance                             | Clear                        |
| Odor                                   | sweet                        |
| Odor Threshold                         | No information available     |
| pH                                     | No information available     |
| Melting Point/Range                    | -37 °C / -34.6 °F            |
| Boiling Point/Range                    | 194 °C / 381.2 °F @ 760 mmHg |
| Flash Point                            | 74 °C / 165.2 °F             |
| Evaporation Rate                       | No information available     |
| Flammability (solid,gas)               | Not applicable               |
| Flammability or explosive limits       |                              |
| Upper                                  | 22.00%                       |
| Lower                                  | 1.10%                        |
| Vapor Pressure                         | 0.4 hPa @ 20 °C              |
| Vapor Density                          | 4.4                          |
| Specific Gravity                       | 1.050                        |
| Solubility                             | practically insoluble        |
| Partition coefficient; n-octanol/water | No data available            |
| Autoignition Temperature               | 498 °C / 928.4 °F            |
| Decomposition Temperature              | > 250°C                      |
| Viscosity                              | No information available     |
| Molecular Formula                      | C8 H8 O                      |
| Molecular Weight                       | 120.15                       |

## 10. Stability and reactivity

|                                  |                                                                                                       |
|----------------------------------|-------------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                                            |
| Stability                        | Stable under normal conditions.                                                                       |
| Conditions to Avoid              | Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition. |
| Incompatible Materials           | Strong oxidizing agents                                                                               |
| Hazardous Decomposition Products | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> )                                               |
| Hazardous Polymerization         | Polymerization can occur.                                                                             |
| Hazardous Reactions              | None under normal processing.                                                                         |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

#### Component Information

| Component        | LD50 Oral         | LD50 Dermal           | LC50 Inhalation    |
|------------------|-------------------|-----------------------|--------------------|
| Oxirane, phenyl- | 4290 g/kg ( Rat ) | 1060 mg/kg ( Rabbit ) | >500 ppm/4h (Rat ) |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | No information available                                                                 |
| Sensitization   | May cause sensitization by skin contact                                                  |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component        | CAS-No  | IARC     | NTP                    | ACGIH | OSHA | Mexico     |
|------------------|---------|----------|------------------------|-------|------|------------|
| Oxirane, phenyl- | 96-09-3 | Group 2A | Reasonably Anticipated | A3    | X    | Not listed |

*IARC (International Agency for Research on Cancer)**IARC (International Agency for Research on Cancer)**Group 1 - Carcinogenic to Humans**Group 2A - Probably Carcinogenic to Humans**Group 2B - Possibly Carcinogenic to Humans**NTP: (National Toxicity Program)**NTP: (National Toxicity Program)**Known - Known Carcinogen**Reasonably Anticipated - Reasonably Anticipated to be a Human Carcinogen*

|                                                   |                                                                                                                                                                                                                                                                               |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Mutagenic Effects</b>                          | Ames test: positive.                                                                                                                                                                                                                                                          |
| <b>Reproductive Effects</b>                       | No information available.                                                                                                                                                                                                                                                     |
| <b>Developmental Effects</b>                      | No information available.                                                                                                                                                                                                                                                     |
| <b>Teratogenicity</b>                             | No information available.                                                                                                                                                                                                                                                     |
| <b>STOT - single exposure</b>                     | None known                                                                                                                                                                                                                                                                    |
| <b>STOT - repeated exposure</b>                   | None known                                                                                                                                                                                                                                                                    |
| <b>Aspiration hazard</b>                          | No information available                                                                                                                                                                                                                                                      |
| <b>Symptoms / effects, both acute and delayed</b> | Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting:<br>Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |
| <b>Endocrine Disruptor Information</b>            | No information available                                                                                                                                                                                                                                                      |
| <b>Other Adverse Effects</b>                      | See actual entry in RTECS for complete information.                                                                                                                                                                                                                           |

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

| Component        | Freshwater Algae   | Freshwater Fish                         | Microtox   | Water Flea           |
|------------------|--------------------|-----------------------------------------|------------|----------------------|
| Oxirane, phenyl- | EC50 = 32 mg/L/96h | LC50 = 34 mg/L/96h<br>(Cyprinus Carpio) | Not listed | EC50 = 11.6 mg/L/48h |

**Persistence and Degradability** Persistence is unlikely based on information available.**Bioaccumulation/ Accumulation** No information available.**Mobility** Will likely be mobile in the environment due to its water solubility.

| Component        | log Pow |
|------------------|---------|
| Oxirane, phenyl- | 1.61    |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

|                             |                               |
|-----------------------------|-------------------------------|
| <b>UN-No</b>                | UN2810                        |
| <b>Proper Shipping Name</b> | Toxic liquid, organic, n.o.s. |
| <b>Technical Name</b>       | Styrene oxide                 |
| <b>Hazard Class</b>         | 6.1                           |
| <b>Packing Group</b>        | III                           |

### TDG

|                             |                               |
|-----------------------------|-------------------------------|
| <b>UN-No</b>                | UN2810                        |
| <b>Proper Shipping Name</b> | Toxic liquid, organic, n.o.s. |

|                             |                               |
|-----------------------------|-------------------------------|
| <b>Hazard Class</b>         | 6.1                           |
| <b>Packing Group</b>        | III                           |
| <b>IATA</b>                 |                               |
| <b>UN-No</b>                | UN2810                        |
| <b>Proper Shipping Name</b> | Toxic liquid, organic, n.o.s. |
| <b>Hazard Class</b>         | 6.1                           |
| <b>Packing Group</b>        | III                           |
| <b>IMDG/IMO</b>             |                               |
| <b>UN-No</b>                | UN2810                        |
| <b>Proper Shipping Name</b> | Toxic liquid, organic, n.o.s. |
| <b>Hazard Class</b>         | 6.1                           |
| <b>Packing Group</b>        | III                           |

## 15. Regulatory information

### International Inventories

| Component        | CAS-No  | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|------------------|---------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Oxirane, phenyl- | 96-09-3 | X   | -    | X    | ACTIVE                                        | 202-476-7 | -      | -   |

| Component        | CAS-No  | IECSC | KECL     | ENCs | ISHL | TCSI | AICS | NZIoC | PICCS |
|------------------|---------|-------|----------|------|------|------|------|-------|-------|
| Oxirane, phenyl- | 96-09-3 | X     | KE-28409 | X    | X    | X    | X    | X     | X     |

#### Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCs - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

### Other International Regulations

#### Authorisation/Restrictions according to EU REACH

| Component        | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                      | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Oxirane, phenyl- | -                                                                   | Use restricted. See item 28. (see link for restriction details)<br>Use restricted. See item 75. (see link for restriction details) | -                                                                                                     |

#### REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component | CAS-No | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous |
|-----------|--------|----------|------------------------------|---------------------------|--------------------------|
|-----------|--------|----------|------------------------------|---------------------------|--------------------------|

|                  |               |                                                                                                  |                                                                                                 |                                   |                                           |
|------------------|---------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-----------------------------------|-------------------------------------------|
|                  |               |                                                                                                  |                                                                                                 |                                   | <b>Substances (RoHS)</b>                  |
| Oxirane, phenyl- | 96-09-3       | Not applicable                                                                                   | Not applicable                                                                                  | Not applicable                    | Not applicable                            |
| <b>Component</b> | <b>CAS-No</b> | <b>Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification</b> | <b>Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements</b> | <b>Rotterdam Convention (PIC)</b> | <b>Basel Convention (Hazardous Waste)</b> |
| Oxirane, phenyl- | 96-09-3       | Not applicable                                                                                   | Not applicable                                                                                  | Not applicable                    | Not applicable                            |

## 16. Other information

**Prepared By**

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www.thermofisher.com

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**Revision Summary**

New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**